Geophysical and Remote Sensing Monitoring of a Snow Patch System in Barton Peninsula Shows Impacts of Warming on Low-Altitude Permafrost

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***Abstract*—Antarctica, a critical regulator of global climate, faces threats to its permafrost and ecosystems from recent warming. However, a quantitative understanding of subsurface responses remains limited, hindering accurate environmental modeling. This gap hinders accurate modeling of future environmental changes. This study investigates the influence of rising air temperatures on thaw depth and permafrost characteristics by quantifying the links between surface environmental changes and subsurface responses. From 2018 to 2024, we integrated meteorological observations, drone and satellite remote sensing, and geophysical surveys—electrical resistivity tomography (ERT) and ground-penetrating radar (GPR)—to assess atmosphere, surface, and subsurface changes. Our results revealed an overall warming trend during the study period, with the average annual air temperature rising by approximately 1 °C and the thaw season extending by up to 50 days. Earlier snowmelt reduced albedo, increasing soil heat absorption and meltwater infiltration. The thaw depth thickened from 1.1 to 1.5 m (maximum) and from 0.65 to 0.85 m (dry sites). ERT indicated reduced resistivity at ~1-m depth, reflecting permafrost ice melt, and localized meltwater pooling at ~3-m depth. Normalized difference vegetation index data showed increased vegetation activity. Our study shows that even slight warming can drive linked physical and ecological shifts in Antarctica, with implications for global climate feedbacks. Our quantitative analysis of the increasing late-summer thaw depth provides important data that can contribute to the validation and improvement of regional climate models.**

***Index Terms*—Antarctica, climate change, geophysical monitoring, permafrost, thaw depth.**

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I. Introduction

ANTARCTICA, which covers approximately 10% of the Earth's surface, is a key regulator of the planet's climate, with approximately 99% of its area covered by snow and ice [1], [2]. This vast natural laboratory holds crucial clues to many scientific challenges facing humanity, from ice cores that preserve records of Earth's past climate to indicators that can help predict future sea level changes [3], [4]. Early Antarctic research primarily focused on relatively accessible coastal areas, where scientific stations from various countries served as research outposts [5]. Recent breakthroughs in survey technology have enabled the study of the Antarctic interior, which is covered by ice sheets several kilometers thick [6]. In particular, the expanding application of geophysical survey techniques that investigate subsurface characteristics in a nondestructive manner has significantly deepened our understanding of underground environments. The data collected using these methods provide a foundation for understanding Antarctica's past and present, as well as predicting future environmental changes [7], [8].

Earth is currently experiencing unprecedented climate change-driven impacts, including rapid Himalayan glacier melting that triggers new floods, increasingly powerful tropical cyclones that cause widespread loss of life and property, and rising sea levels that endanger low-lying countries by placing their populations, infrastructure, and economies at severe risk [9], [10], [11], [12]. However, the most dramatic manifestations of these impacts and clear warnings about future changes are evident in polar regions. For example, in the Arctic, permafrost thaw is accelerating, causing ground collapse and increasing the risk of unpredictable secondary and tertiary hazards such as the reactivation of ancient microorganisms and the release of large quantities of long-trapped methane gas [13], [14], [15]. While these processes are actively studied in the Arctic, similar permafrost-related changes are likely to occur in Antarctica, particularly in the relatively warmer Antarctic Peninsula and coastal regions [16], [17]. However, the rate

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and scale of Antarctic permafrost change and their specific effects on the continent's characteristic ground stability and ecosystems remain poorly understood.

The Antarctic continent is not immune to climate change, and dramatic environmental shifts have been reported in recent years. In 2020, temperatures exceeding 20 °C were recorded for the first time on Seymour Island (64°14′S, 56°37′W), which is home to an Argentinian research station. On the Barton Peninsula (62°14′S, 58°46′W) of King George Island, our study area, the average annual temperature has risen steadily since the 1950s, despite intermittent warming and cooling periods [18]. This warming is driving ecosystem changes, including the rapid retreat of nearby ice shelves, declines in penguin populations, and the spread of flowering plants, such as *Deschampsia antarctica* [19].

To study these changes, researchers have combined satellite remote sensing, which covers a wide area, with geophysical surveys, which probe subsurface conditions. Satellite data are valuable for tracking large-scale changes such as glacial retreat and sea surface temperature; however, 2-D surface information alone cannot fully resolve the 3-D subsurface processes that drive these changes. To overcome this limitation, geophysical methods have been applied in precision surveys. Electrical resistivity tomography (ERT) maps the permafrost and moisture distribution by exploiting the differences in physical properties of ice and water [20], [21], [22], while ground-penetrating radar (GPR) measures the thickness of the active layer and glaciers [23], [24], [25]. These high-resolution datasets help to overcome the logistical constraints of Antarctic fieldwork and improve the accuracy of environmental models in predicting future changes. Therefore, the application of geophysical survey techniques is essential for investigating and understanding dynamic subsurface changes over time and space.

In this study, we hypothesized that an increase in temperature alters the heat transfer to the subsurface, leading to physical changes in the thaw depth and permafrost. We comprehensively analyzed ground observations, remote sensing data, and geophysical survey results to investigate the impact of climate change on the study area from multiple perspectives.

II. Study Area and Environmental Changes

A. Study Area

The study area [see Fig. 1(a)] is located on the Barton Peninsula, King George Island, Antarctica, at 62°14′ 24″S, 58°44′ 34″W, about 3 km southeast of King Sejong Station (62°13′ 22″S, 58°47′ 23″W), a scientific research facility operated by the Republic of Korea. Approximately 25 m north of the site is a gentle hill that is approximately 50 m high. Drone-derived digital elevation model (DEM) data indicate that the main survey section has an average elevation of ~31 m, with a relief of approximately 3 m. Fig. 1(b) shows a drone image of a 40-m survey transect within the study area, revealing distinct vegetation changes along its length.

The Barton Peninsula, where the study area is located, features a geological setting of coexisting andesitic bedrock and Quaternary–Holocene marine sediments. Dominant lithologies include agglomerate, breccia, and quartz diorite, with reported densities of 2.5, 2.6, and 2.7 g/cm^3, respectively. The average density of the shallow subsurface is approximately 2.2 g/cm^3 [26], [27]. The hydrological conditions of the site are strongly influenced by snow accumulation on the northern hills. Snow persists through summer as patches begin to melt in October when temperatures rise above freezing. The resulting meltwater flowed downslope, sustaining the pond at a relatively low elevation. The pond measured approximately 100 m along its long axis and 40 m along its short axis.

This moisture gradient from the northern hill produced distinct vegetation zones in the study area. *Sanionia uncinata*, a moss adapted to moist environments, dominates the lower slopes of hills where the water supply is the highest, and its color deepens to dark green as moisture increases. Moving south from the downhill slope to progressively drier areas, the abundance of *S. uncinata* gradually decreased, giving way to a 10-m wideband where the lichens *Ochrolechia frigida* and *Psoroma* spp. coexist. In the drier soils to the southwest (SW), where the elevation increases slightly, the lichen communities are dominated by *Cladonia* spp. and *Usnea antarctica*.

B. Environmental Changes

During the study period, distinct changes were observed in the snow patch on the northern slope, which is the main source of water in the study area. Fig. 2 shows drone orthophotos of the same area collected in early February 2018 and mid-February 2024. In 2018 [see Fig. 2(a)], the slope and surrounding ponds were mostly covered by snow. In the summer of 2024 [see Fig. 2(b)], most of the snow around the pond disappears, and the bedrock is widely exposed in the central part of the slope. These changes suggest that, during the study period, summer snowmelt exceeded winter snow accumulation, resulting in a significant reduction in the overall volume of the snow patch.

To analyze changes in air temperature in the study area, we used data collected at 10-min intervals (sensor height: 1.8 m) from January 2018 to April 2024 from an automatic weather station (AWS) at the nearby King Sejong Station. The annual average temperature increased from −1.8 °C in 2018 to −0.59 °C in 2021 and then to −0.93 °C in 2023. Overall, the annual average temperature rose by approximately 0.17 °C during the study period, a trend that likely contributed to the decrease in the snow patch observed in Fig. 2.

Fig. 3 shows the detailed changes in air and soil temperatures. A soil temperature sensor was installed approximately 10 m east of the center of the survey section. The onset of soil thawing between 2019 and 2024 occurred earlier than that in 2018, with the earliest thawing observed during 2021–2022. Over time, soil temperature increased above 0 °C.

The onset of the thaw season occurred progressively earlier each year, with the earliest onset observed in 2021–2022. Over time, soil temperature increased above 0 °C. The onset of the thaw season occurred progressively earlier each year, with the earliest onset observed in 2021–2022. Soil freezing began after April 15 in all years. As shown in the continuous time-series data (see Fig. 3), soil temperature at a depth of 5 cm showed a strong correlation with air temperature but with

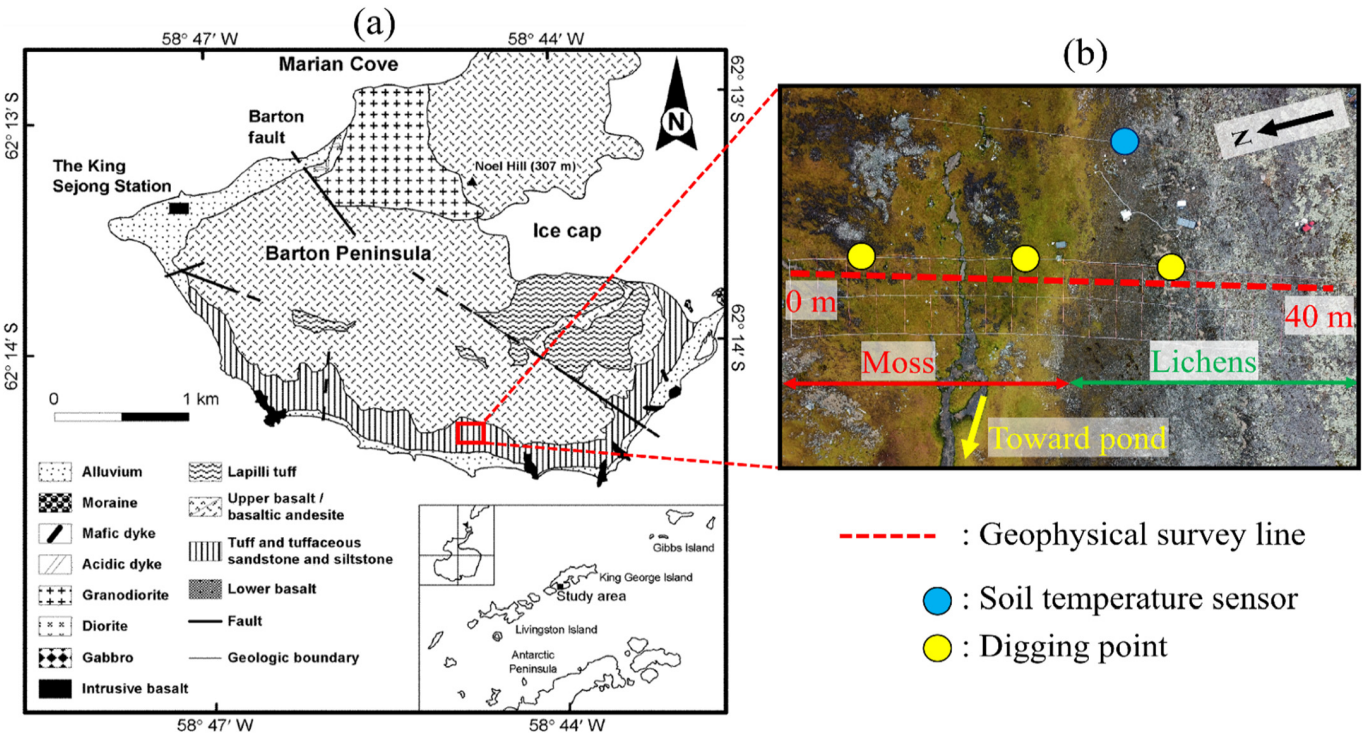


Fig. 1. (a) Geological map (reproduced from Kim et al. [28]) and (b) aerial drone photograph of the surveying site in Barton Peninsula, King George Island, Antarctica. Sky blue dot indicates soil temperature sensors, and yellow dots mark pit excavation survey locations.

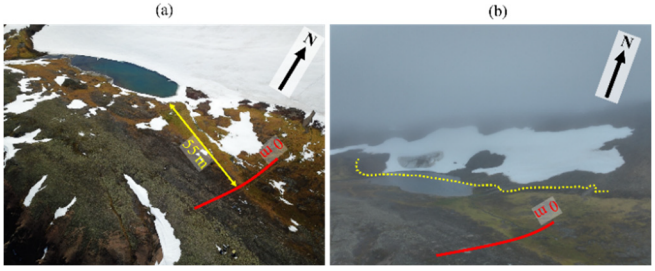


Fig. 2. Aerial drone photographs of snow cover conditions in the study area. (a) Early February 2018 and (b) mid-February 2024. The yellow dashed line represents the snow boundary in early February 2018. The solid RED line represents the survey line. The survey line was located 55 m from the pond.

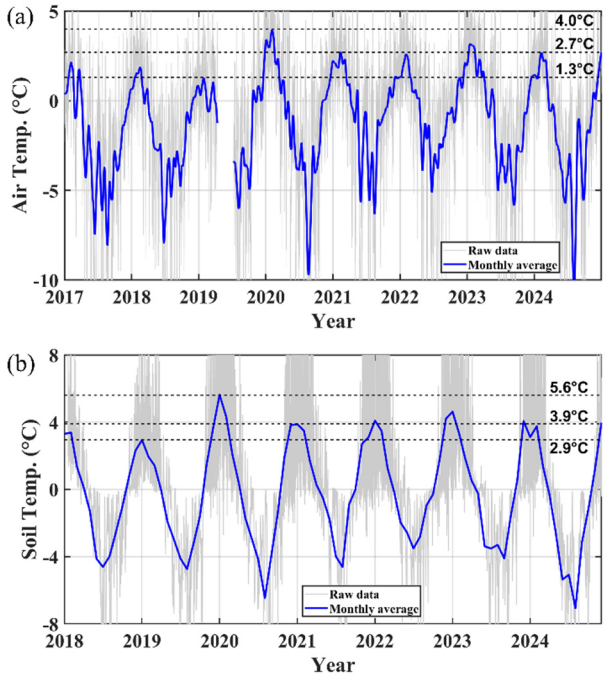


Fig. 3. (a) Temporal variations in air temperature (1.5 m above ground) and (b) soil temperature (5-cm depth). Air temperature data were unavailable from late April to late June 2019 due to a malfunction of the AWS.

significant seasonal differences. These data clearly demonstrate the interannual variability in the timing and duration of surface freezing and thawing cycles. During the summer thaw season, soil temperatures were consistently higher than air temperatures although the magnitude of this difference exhibited significant interannual variability. This temperature relationship is highly dependent on the insulating effect of snow cover, which keeps soil temperatures significantly milder than air temperatures during winter. We also observed that the maximum soil temperatures were typically delayed by several days compared with the air temperature peaks, reflecting slower heat conduction in the soil [29], [30]. Overall, the data indicate continued warming during the study period, particularly in terms of the timing and duration of surface freezing and thawing.

To examine ground surface changes over seven years, we analyzed snow cover and NDVI in the study area using Korea Multipurpose Satellite-3A (KOMPSAT-3A, also known as Arirang 3AA satellite) data (see Fig. 4). NDVI, developed in the early 1970s, quantifies vegetation by comparing reflectance in the near-infrared (NIR) band (0.86 μm, strongly reflected

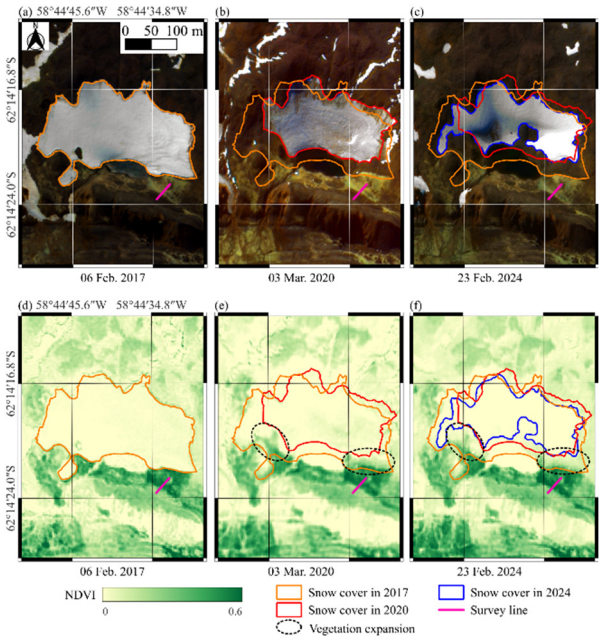


Fig. 4. Changes in snow cover distribution and NDVI in the study area from KOMPSAT-3A satellite images. The top row (a)–(c) shows snow cover in 2017, 2020, and 2024, respectively. The bottom row (d)–(f) shows NDVI values for the same periods. Data for 2018 were unavailable due to cloud cover; 2017 data were used as a proxy for the initial state. The black ellipse highlights the area where vegetation distribution has expanded.

by vegetation) and the red (RED) band (0.66 μm, strongly absorbed by vegetation) [31], [32]

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}} \tag{1}$$

where NIR and RED represent the spectral reflectance of their respective bands. High vegetation vitality or density increases the NIR reflectance and RED absorption. NDVI values range from –1 to 1, with values near 1 indicating active vegetation growth and values near 0 indicating bare ground or rock. We examined NDVI changes in February 2017, March 2020, and February 2024. Data from 2018 were unavailable because of persistent cloud cover. Therefore, to establish a reasonable baseline for our analysis, we used the data from February 2017 as a proxy for the initial state. A climatic comparison was conducted to validate the proxy. As shown in Fig. 3, the continuous air temperature data from the nearby King Sejong Station showed highly comparable summer temperature patterns for 2017 and 2018, with both years exhibiting similar summer (January–February) monthly average peaks. We selected the KOMPSAT-3A operated by Korea Aerospace Research Institute (KARI) [33] as its 2.2-m multispectral resolution provides a higher spatial resolution than other available satellite data, making it more advantageous for analyzing the fine-scale vegetation patterns within our small study site. In addition, these satellite data guarantee high quality through atmospheric correction, which is applied before release.

The changes in snow cover [see Fig. 4(a)–(c)] are consistent with the drone observations shown in Fig. 2. In February 2018, snow was widely distributed on the slope; however, it

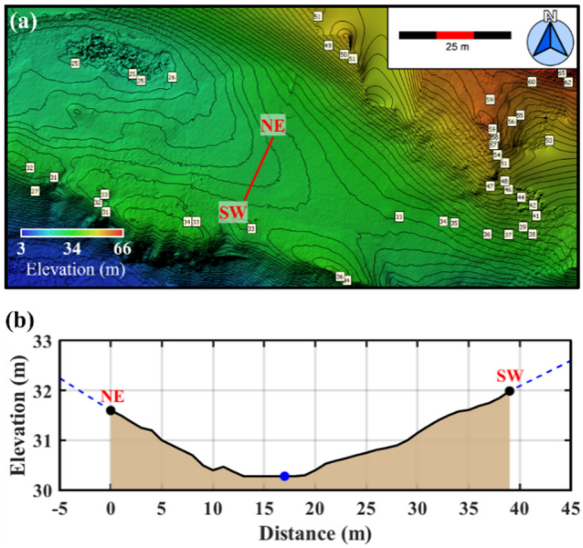


Fig. 5. (a) DEM data and (b) elevation profile of the geophysical survey line (black solid line). The blue dot indicates the lowest elevation point, and the magenta triangles represent 40 electrodes (1-m intervals). Black dots mark the endpoints of the survey line.

melted earlier and more rapidly. By February 2024, most snow cover had disappeared from the lower slope and around the pond. This reduction in snow cover directly affected vegetation distribution [see Fig. 4(d)–(f)].

In 2017, the NDVI values were close to zero in the snow-covered and bare-ground areas. By 2020, with higher temperatures and extensive snowmelt, the NDVI data showed a general expansion of vegetation within the melted areas compared to 2017. The 2024 data indicated a slightly further expansion of vegetation activity compared to that in 2020, during which temperatures and snowmelt extents were similar. Notably, NDVI increases were greatest in low-lying areas with a reliable meltwater supply (southward direction) compared to higher, drier areas (northward direction).

In summary, drone and satellite observations, along with the air and soil temperature analyses presented in Section II, demonstrated continued warming in the study area over the past seven years. This warming has driven distinct changes in the surface environment, including reduced summer snow cover and increased vegetative activity. These surface changes provide direct evidence of physical changes occurring in the subsurface. Understanding the relationship between these surface changes and subsurface processes requires quantitative analysis using geophysical surveying techniques.

III. Variation of Subsurface Conditions

In this study, we employed geophysical surveys, a non-destructive technique, to quantitatively analyze the effects of climate change on the physical state of the subsurface. To visualize changes in water content during the permafrost freeze–thaw process, we applied ERT and GPR, both of which are sensitive to subsurface water distribution and provide high resolution in the shallow subsurface.

Fig. 5(a) shows a DEM of the study area, revealing a basin-shaped topography. A hill with an elevation exceeding 60 m

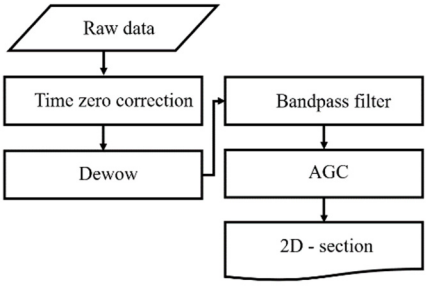


Fig. 6. Major data processing steps applied to GPR raw data.

is located approximately 50 m to the northeast (NE) and serves as the main source of meltwater runoff during snowmelt periods. In contrast, a cliff lies approximately 20 m to the SW. Fig. 5(b) shows the elevation profile along the ERT survey line. The profile formed a gentle trough, with both the NE and SW endpoints, situated at an elevation of approximately 32 m and descending to the lowest point of approximately 30 m near the 17-m mark. The total elevation difference (relief) along the profile is approximately 1.7 m. We conducted four repeated surveys along the same 40-m (based on identical starting and ending points) survey lines at two-year intervals from February 2018 to February 2024. Data acquisition was scheduled for mid-February, the peak ice-melting period in Antarctica, except in 2022, when data were collected in late January owing to field conditions.

The ERT survey used 40 electrodes at 1-m intervals. We applied the Wenner array, which is widely used in shallow geological investigations because of its excellent vertical resolution and sensitivity to horizontal structural changes [34].

The data were processed using Res2dinv, a reliable 2-D electrical resistivity inversion software developed by Aarhus Software (www.aarhusgeosoftware.dk/res2dinv). Res2dinv uses the widely adopted smoothness-constrained least-squares inversion method, which iteratively determines a subsurface resistivity model that minimizes the following objective function during the inversion process [35].

Producing high-resolution images of the boundary between the thaw depth and the permafrost layer is essential for precisely tracking changes in thaw depth. To this end, GPR surveys were conducted, which offer a high vertical resolution suitable for boundary detection. Data were collected at 0.1-m intervals using a 500-MHz frequency antenna and processed using ReflexW software (www.sandmeier-geo.de).

To improve the signal-to-noise ratio of the raw GPR data, the following processing steps were applied (see Fig. 6). First, an appropriate signal penetration range was set by applying a time-zero correction to the initial movement of the signal and defining the time window. The time window was fixed at 50 ns for proper interpretation of the GPR data. Next, we applied a dewow filter to remove the low-frequency "wow" noise generated by the system. Bandpass filters (50/100/550/600 MHz) were then used to effectively suppress both field and residual noise. Finally, automatic gain control was applied to produce the final GPR cross-sectional diagram, compensating for the amplitude reduction over time due to radio-wave attenuation and amplifying signals in the later time period.

A common-midpoint survey was conducted at a 35-m point on the survey line to determine the radio-wave velocity required for depth conversion. The analysis yielded an average velocity of 0.08 m/ns, which was applied uniformly to all GPR cross sections. The 500-MHz antenna provides a high theoretical vertical resolution (approximately 4–5 cm based on $\lambda/4$ at 0.08 m/ns) [36], which is sufficient for resolving changes at the 5–10-cm scale.

Fig. 7 presents four ERT cross sections measured at two-year intervals. The 2018 cross section, which represents the initial state [see Fig. 7(a)], shows a complex resistivity distribution. In the 3–18-m segment near the start of the survey line, the surface layer exhibits a low resistivity of 200 Ωm or less. However, a relatively high-resistivity band (approximately 10 kΩm) was observed at a depth of about 1 m in the 3–12-m section, suggesting abundant ice within the voids. In contrast, the 12–20-m section displayed a uniformly low resistivity, interpreted as the penetration of surface meltwater, as confirmed by on-site pit excavation. Toward the latter part of the line (26–35 m), higher resistivity values (around 4–5 kΩm) suggest less influence from slope-derived meltwater.

The 2020 and 2022 results [see Fig. 7(b) and (c)] show a clear decreasing trend in the overall electrical resistivity, particularly in the 4–12-m section. In the 15–18-m section, a low resistivity (≤400 Ωm) extends to a depth of 5 m, indicating an increase in groundwater infiltration. These changes strongly correlated with the increase in average annual temperature and the number of days with above-freezing weather (average daily temperature > 0 °C) during this period. That is, warming resulted in an earlier thawing season and increased meltwater, which accelerated the thawing of the permafrost table. However, a localized increase in electrical resistivity was observed in the 2022 data between the 24- and 32-m marks. This anomaly is likely attributable to the survey timing. As stated in our methodology, the 2022 survey was conducted in late January, approximately three weeks earlier than in other years. It is plausible that this specific section, which is topographically higher and drier, had not yet fully thawed by late January, resulting in higher electrical resistivity values compared with the mid-February state observed in 2020.

In contrast, the 2024 cross section [see Fig. 7(d)] exhibited the opposite pattern, with a slight overall increase in electrical resistivity values compared to 2022. This increase is directly related to the year-on-year decrease in the average annual temperature and the number of days with above-freezing weather in 2023, which was the only year in the last five years with a temperature drop. This suggests that a single year of cooling has an immediate impact, promoting refreezing and slowing the thaw depth and permafrost table.

Fig. 8 presents the GPR cross sections acquired along the same survey line and at the same time as the ERT survey, quantitatively illustrating the changes in thaw depth over time. In each cross section, the red dashed line marks the boundary between the top of the upper permafrost and the thaw depth, the sky-blue dashed line marks the organic layer, and the solid red arrow at the top indicates the section with the most pronounced signal attenuation.

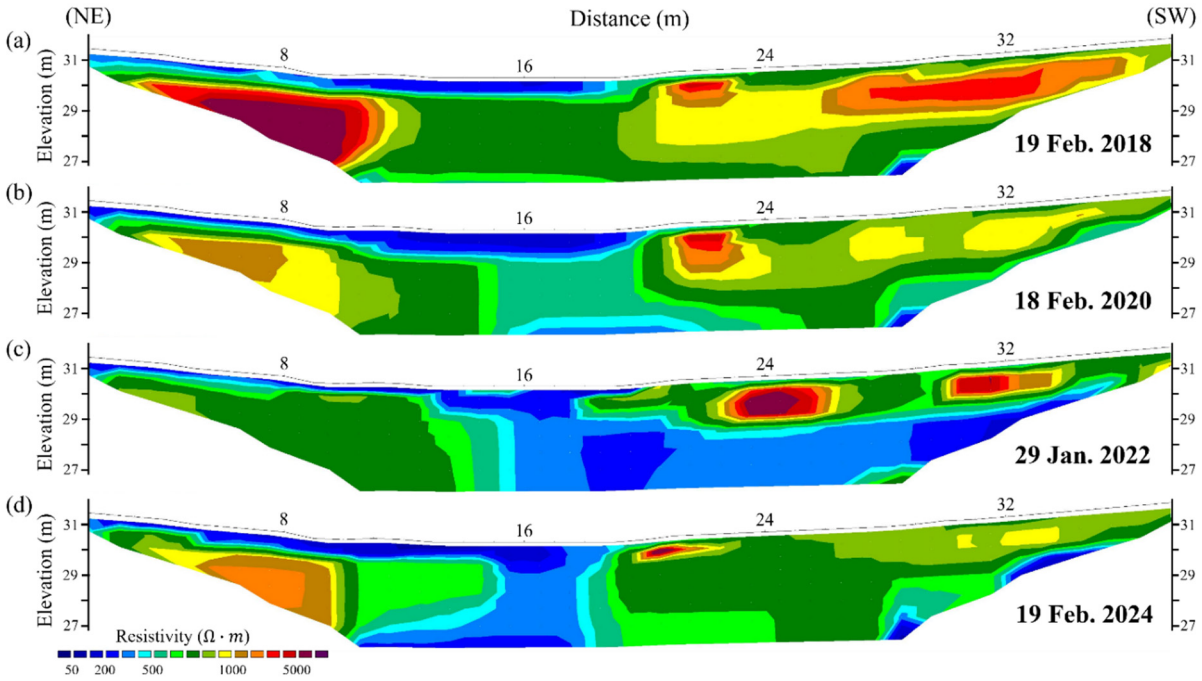


Fig. 7. 3-D inversion cross sections from ERT conducted in (a) 2018, (b) 2020, (c) 2022, and (d) 2024.

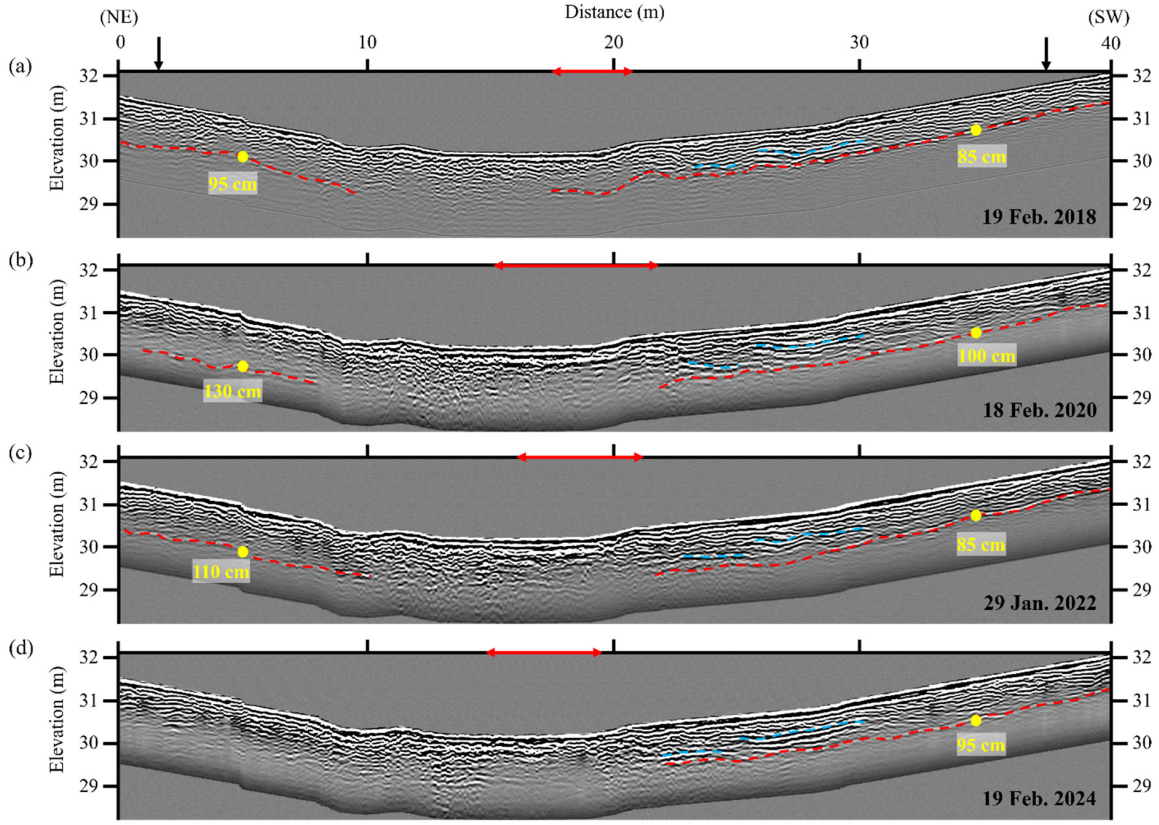


Fig. 8. GPR cross sections showing changes in the thaw depth from (a) 2018 to (d) 2024. The red dashed line marks the lower boundary of the thaw depth, and the sky-blue dashed line represents the boundary of the surface organic layer. The solid red arrow at the top of each profile indicates the section with the most severe GPR signal attenuation (15–22 m), corresponding to the zone of the lowest electrical resistivity in the ERT results. The black arrows on the top profile indicate the start and end points of the ERT survey line shown in Fig. 7.

The physical properties of the cross section varied distinctly along the survey line. The first half (0–20 m) shows shallower penetration depths than the second half (20–40 m) because of the severe diffraction and signal attenuation. The

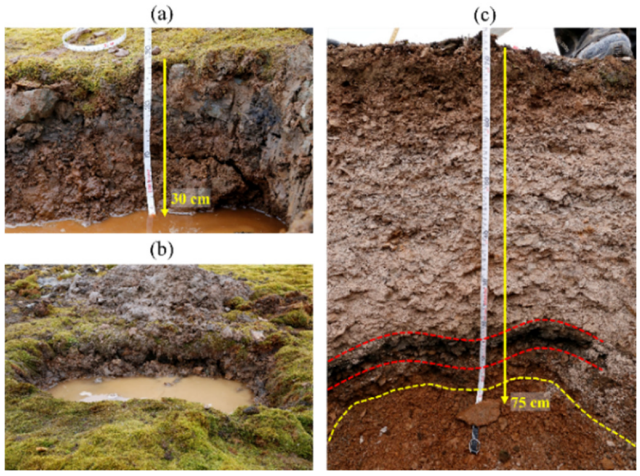


Fig. 9. Pit excavation surveys at three points (5, 17, and 28 m) along the survey line. (a) Soil cross section at the 5-m point. (b) 17-m point, where the surface layer is fully saturated. (c) Detailed soil cross section at the 28-m point. The solid RED line marks the upper and lower boundaries of the high-moisture black organic layer. The yellow dashed line marks the lower boundary of the thaw depth of 75 cm.

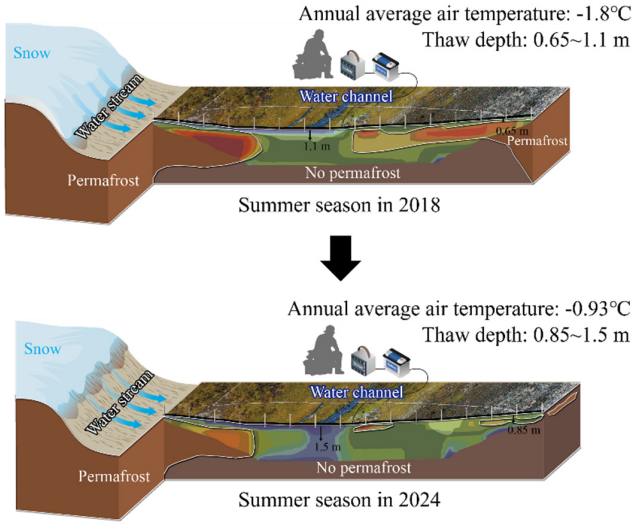


Fig. 10. Schematic of major environmental changes in the study area between summer 2018 and summer 2024. Over the past six years, snow patches have retreated, and the thaw depth has increased from 0.65–1.1 to 0.85–1.5 m due to meltwater inflow.

15–22-m section, marked by the solid red arrow, corresponds to the zone with the lowest electrical resistivity in the ERT results. The analysis indicated that the electromagnetic wave energy loss was the greatest in this section because of its high water content and the presence of a gravel layer.

Changes in the thaw depth over time are clearly visible from the depth variations in the red dashed line. In 2018 [see Fig. 8(a)], the lower boundary of the thaw depth was at a depth of 0.65–1.1 m. By 2020 [see Fig. 8(b)], during a warmer summer, it had increased by approximately 0.2 m overall. In 2022 [see Fig. 8(c)], despite measurements being taken three weeks earlier than in 2018, the lower limit was still 5–10 cm deeper than in 2020. In contrast, 2024 [see Fig. 8(d)], following a relatively cool summer, did not show a further increase in the thaw depth compared to 2020. These year-to-year depth variations in the red dashed line provide direct evidence that the thaw depth is highly sensitive to summer temperatures.

To improve the reliability of the geophysical survey results, we performed pit excavation surveys at three points (5, 17, and 28 m) that exhibited distinctly different physical characteristics (see Fig. 9). At the 5-m point, located at a relatively high elevation, groundwater was observed at a depth of 0.25 m within a mixture of gravel (5–10 cm in diameter) and viscous mud [see Fig. 9(a)]. At the 17-m point, which had the lowest elevation, the ground was fully saturated, with water pooling on the surface when stepped on [see Fig. 9(b)]. Field observations confirmed that a high water content directly caused severe GPR signal attenuation and low electrical resistivity.

At the 28-m point, located at the highest elevation and in a relatively dry area, the cross section revealed a distinct soil structure [see Fig. 9(c)]. Beneath the 10-cm-thick organic surface layer was a dry layer of mixed sand and small stone. At a depth of 0.6 m, a 10-cm-thick black organic layer with high moisture content reappeared. The solid red line in Fig. 9(c) indicates the upper and lower boundaries of the layer. Below it, at a depth of 0.75 m, we observe the thaw depth, corresponding to the yellow dashed line. This 0.75-m depth exactly matched the thaw depth determined from the 2018 GPR cross section, directly validating the geophysical survey and analysis results.

The thicker thaw depth in the first half (0–20 m) of the survey line can be explained by differences in soil thermal conductivity. The thermal conductivity of water is approximately 22 times higher than that of air [37]; therefore, moisture-rich soil, such as at the 17-m point, transfers heat to the subsurface much more efficiently than the dry soil at the 28-m point. In the first half of the survey line, where meltwater from the slope was concentrated, more thermal energy reached the upper part of the permafrost, thereby accelerating thawing. This pattern, commonly observed in the ERT and GPR results over time, clearly shows that water distribution is a key factor in subsurface changes in polar regions.

IV. Discussion and Conclusion

In this study, we analyzed the effects of climate change on the polar environment in a specific area of the Barton Peninsula, King George Island, Antarctica, from 2018 to 2024. We observed air and subsurface temperatures, analyzed surface environments using drones and satellites, and comprehensively tracked subsurface changes using ERT and GPR. The results clearly show that over the past six years, the study area has undergone a series of surface changes triggered by rising temperatures.

The environmental changes observed in this study demonstrate a clear causal relationship, indicating that the changes began with an increase in air temperature. First, the thermal environments of the air and ground surfaces changed. From 2018 to 2023, the average annual temperature in the study area increased by approximately 1 °C, and the summer period (average daily temperature > 0 °C) lengthened by about 50 days. Consequently, as confirmed by drone and satellite images, the winter snow cover melted earlier and in greater quantities than in previous years.

Second, these surface changes alter the energy transfer to the subsurface. Early snowmelt reduces the surface albedo and increases solar energy absorption [38]. In addition, the increased volume of meltwater infiltrates the ground with more heat, directly increasing the soil temperature.

Third, these processes physically alter the thaw depth and permafrost. As shown in Fig. 10, the thaw depth increased significantly over the past six years. GPR results show the maximum thaw depth increased by approximately 0.4 m, from 1.1 m in 2018 to 1.5 m in 2024. It should be noted that these surveys, conducted in mid-February (or late January), captured the thaw depth at that specific time. This measurement may not represent the absolute maximum annual thaw (i.e., the active layer thickness), which regional studies suggest often occurs after March. However, the significant and consistent increase in the observed thaw depth over the six-year period provides strong evidence for the hypothesis that the maximum active layer thickness itself is increasing. The thaw depth observed in this study was shallower than that reported for Arctic regions with similar environmental conditions. This difference in thaw depth is presumed to result from the relatively lower summer temperatures in the Antarctic compared with those in the Arctic [39]. Even at a relatively dry location, the thaw depth increased from 0.65 to 0.85 m. ERT results also consistently showed the thaw depth. At a depth of approximately 1 m, interpreted as the thaw depth and permafrost boundary, the electrical resistivity decreased significantly, indicating the thawing of permafrost, increased water content, and a lowered boundary. The low-resistivity anomaly at a depth of approximately 3 m in the topographically low 15–18-m section was interpreted as meltwater accumulation derived from upslope runoff. This persistent anomaly, extending well below the typical thaw depth, suggests the presence of an unfrozen zone (Talik). Accordingly, our ERT data indicate the occurrence of discontinuous permafrost within the study area rather than a spatially continuous permafrost layer. These interpretations are presented in the conceptual model shown in Fig. 10.

Finally, these subsurface changes affect the ecosystem. Earlier snowmelt lengthened the period of soil exposure, raised soil temperatures, and increased vegetation activity, as confirmed by the satellite NDVI data.

The changes in the thaw depth identified in this study have broader engineering and Earth system implications, extending beyond the physical changes in the specific study area. First, as the upper permafrost and ground support weaken, the long-term stability of structures and runways along the Antarctic coast, including King Sejong Station, could be threatened. Second, as the thaw depth deepens, large amounts of organic matter previously trapped in permafrost may begin to decompose due to microbial activity, releasing potent greenhouse gases such as carbon dioxide and methane (CH_4) into the atmosphere [40], [41], [42]. The balance between the carbon sink and release depends on the amount of photosynthesis by plants or other photosynthetic microorganisms and the degradation of organic materials by microbial activity. Previous studies in the Antarctic area have shown that carbon release is the general result of increasing thaw depth [43], [44].

In this study, we successfully integrated meteorological observations (atmospheric), remote sensing (ground surface), and geophysical survey (subsurface) data from a specific Antarctic coastal region. We comprehensively investigated the chain reactions in the environmental system caused by climate change. The academic significance of this study lies in its comprehensive nature. In particular, we quantitatively presented subsurface structural changes, such as variations in thaw depth, providing foundational data that can improve the accuracy of future climate change prediction models.

We relied on 2-D survey line data, which limited our ability to capture the full spatial heterogeneity of subsurface changes. Furthermore, because the data collection was concentrated in the summer, it did not fully reflect the dynamic processes of freezing and thawing throughout the year. Future studies should employ 3-D geophysical surveys to identify the spatial patterns more clearly. Long-term intensive monitoring is also needed for the 15–22-m section, where the most dynamic changes were observed, to determine the core mechanisms driving these changes. Furthermore, high-precision year-round time-series data from autonomous observation systems should be combined with numerical modeling to clarify the quantitative relationship between meltwater inflow and thaw depth changes. This will advance the predictive research on future changes.

In conclusion, we have confirmed that the Barton Peninsula in Antarctica has experienced complex and accelerated chain reactions involving snow, soil, permafrost, and vegetation over the past six years, despite an average annual temperature increase of only about 1 °C. This demonstrates that polar environments are highly sensitive to small temperature changes and underscores the need for ongoing precise monitoring.

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